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Preparation and characterization of ZnO-PP nanocomposite fibers and non-woven fabrics

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ABSTRACT

In this work, zinc oxide nanorods are synthesized by polyvinyl alcohol-based sol-gel Pyrolysis method and also applied to prepare the antibacterial ZnO/polypropylene nanocomposite fibers and non-woven fabrics. The fibers are produced by the melt spinning and the nonwoven is produced by thermo banding method. The samples are characterized by SEM and TEM. The antibacterial activity of the fibers is qualitative assessed by the agar diffusion and parallel streak and quantitative test methods for percentage reduction of bacteria colonies. The topographical analysis of the treated fiber and untreated fiber are studied and compared. The results show that the treated fibers and non-woven fabrics have the significant antibacterial and antiallergic properties. In addition, the samples show good mechanical strength and UV protection.

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KEYWORDS

Nanocomposite; nanofiber; nonwoven; zinc oxide nanorod; antibacterial; antiallergic

1. Introduction

Health and hygienic properties are the primary requirements for human beings to live comfortably. The defense clothing system has very high potential for cross infections because they are living in groups in various camps and in extreme climates bed linen, socks, wool blankets and clothing are the initial carriers of microbes and hence cross infection are possible.

Inorganic materials such as metals and metal oxides have been attracting a lot of attention over the past decade due to their abilities to withstand harsh process conditions (Fu et al., 2005; Makhluף et al., 2005). Metal oxides such as TiO₂, ZnO, MgO, and CaO are the particular interest as they are not only stable under harsh process conditions, but also generally regarded as a safe material to human beings and animals (Stoimenov, Klinger, Marchin, & Klabunde, 2002).

Zinc oxide has recently special achieved attention regarding potential electronic applications due to its unique optical, electrical, and chemical properties (Baxter & Aydil, 2005). The availability of the wide range of nanostructures makes ZnO as an ideal material for nanoscale optoelectronic (Luo, 2009), a piezoelectric nano generator (Song, Zhou, & Wang, 2006), as well as an efficient material for biotechnology (Wang, 2004). Furthermore, ZnO appears to be strong resisted to microorganisms (Sawai et al., 2000). There are some reports (Sawai & Yoshikawa, 2004) on the considerable antibacterial activity of CaO, MgO, and ZnO which is attributed to the generation of reactive oxygen species on the surface of these oxides.

The advantage of using this inorganic oxide as an antimicrobial agent is that they contain environmentally safe mineral elements essential to humans and exhibit strong activity even when administered in small amount. The activity is quantitatively evaluated by studying the growth medium caused by the bacterial metabolism (Sawai, Doi, Maekawa, Yoshikawa, & Kojima, 2002). Many researchers have attempted to correlate the biological activity of inorganic antibacterial agents with the size of the constituent particle (Brayner et al., 2006; Stoimenov et al., 2002). On the other hands, ZnO nanoparticles have been extensively studied as an antimicrobial agent due to their photocatalytic under UV light (Chang, Lin, Chen, Wang, & Ho, 2002; Rana, Rawat, Sorensen, & Misra, 2006). Nowadays, there is a much need to develop such as antimicrobial material for PP fiber and non-woven textile that would be useful for health and the environment. Therefore, we emphasized in the synthesis and application of safe & potential Nanomaterial for fiber and textile.

In this study, we have prepared zinc oxide nanorods by the poly vinyl alcohol (PVA)-based gel combustion method. The ZnO nanorods has been used for the preparation of ZnO/PP nanocomposite in fiber and non-woven form.

2. Experimental

2.1. Material and reagents

All material and reagents had high purity and purchased from Merck. Polypropylene (PP) granules were purchased from Navid

Zar Petroleum Co. (Iran). Double-distilled water used in all the experiments.

2.2. Instrumentals

Incubator from TEB Azma Co. (Iran) and autoclave and steroclave from Pars Teb Novin Co. (Iran) were used in thermal treatments. Colony counter from the Teb Azma Co. (Iran) was used for the counting of bacteria colonies in the samples. Pilot plant of melt spinning from the Hergeth Hollingsworth Co., dozing and colortronic equipment from Neumag Co. were used for fiber and nanocomposite fiber production. Universal testing machine from the Lenzing Labor Co. and material profiling machine from the Zwick Co. were used for the physical tests of fiber and tenacity of non-woven fabric, respectively. A scanning electron microscope with EDX instrument from Philips Co. (XL30) was used for the studying of morphology, particle size, and surface analysis of the prepared ZnO samples. AJEOL-200 CX transmission electron microscope was used to obtain TEM images. Shimadzu Double-beam UV-Visible spectrometer was used to determine the ZnO band gap energy.

2.3. Procedures

2.3.1. Preparation of zinc oxide nanorods

In this procedure, 91 g of mixed ethanol: water (70:30) solvent was used to dissolve 6 g zinc acetate, and 3 g PVA. The mixture was heated up to 70 °C to form a homogeneous sol solution. The obtained sol was heated to 90 °C to evaporate the solvent and form a hard homogenous gel. The final gel was pyrolyzed at 400 °C for 4 h. During the pyrolysis process, the PVA polymeric network was slowly burnt through the outer surface and zinc acetate salt was calcinated and converted into ZnO nanorods. The obtained samples were crushed to prepare a fine powder.

The morphology, particle size and chemical composition of the prepared samples was analyzed by SEM, TEM and UV-visible spectrometer.

2.3.2. Preparation of ZnO/PP nano composite fibers and non-woven fabric

The synthesized PP granulate is heated up to 230–250 °C. Appropriate amount of ZnO nanorods was added to the molten polymer (1 %wt). The melted ZnO/PP nanocomposite was extruded to form a homogeneous molten and converted into the ZnO/PP nanocomposite fiber with 0.5 mm average diameter and 3 mm length which used as master batch. The prepared master batch is mixed with PP granules and melted. The molten mixture was extruded in the temperature of 220–250 °C and converted to the final fibers by melt spinning method. The prepared fibers is processed to the nonwoven fabrics by thermobonding method.

2.3.3. Assessment of antibacterial activity

AATCC 100-1993 and AACC 147-1992 test methods and Agar diffusion method were used to evaluate the antibacterial activities of ZnO/PP nanofibers and nonwoven fabrics.

2.3.3.1. Qualitative test

2.3.3.1.1. Agar diffusion method. The Sterile bacteriostasis agar was dispensed in the sterile petri plates. After 24 h, broth cultures of the test organism (*E. coli* and *S. aureus*) were used as inoculums. Using sterile cotton swab, the test organisms were swabbed over the surface of the agar plates. The test fabrics (treated with ZnO Nanorods) and control fabrics (treated with micrometer-sized ZnO and virgin fiber) were gently pressed in the center of mat culture. The plates were incubated at 37 °C for 18–24 h.

2.3.3.1.2. Parallel streak method: (AACC test method 147-1992). In this test exactly similar to Section 2.3.3.1.1, the sterile bacteriostasis agar was dispensed in petri plates. After 24 h, broth cultures of the test organisms (*E. coli* and *S. aureus*) were used as inoculums. Using 2 mm inoculation loop, 1 loop full of culture was loaded and transferred to the surface of the agar plate by making 7.5 cm long parallel streaks 1 cm apart in the center of the plate, without refilling the loop. The test specimens (fabric treated with ZnO nanorods, fabrics treated with micrometer-sized ZnO and virgin fiber) were gently pressed transversely, across the five inoculums of streaks to ensure intimate contact with the agar surface. The plates were incubated at 37 °C for 18–24 h.

2.3.3.2. Quantitative tests

2.3.3.2.1. Percentage reduction test: (AATCC test method 100-1998). Specimens of the test material were shaken in a known concentration of bacterial suspension and the reduction in bacterial activity in standard time was measured. The efficiency of the antimicrobial treatment is determined by comparing the reduction in bacterial concentration of the treated sample with that of control sample expressed as a percentage reduction in standard time. Percentage reduction was calculated using the following formula:

$$R = \frac{(A - B)}{A} \quad (1)$$

Where *R* is reduction percentage, *A* is the number of bacteria in the broth with treated test sample immediately after inoculation i.e. at zero contact time and *B* is the number of bacteria recovered from the broth inoculated with treated test sample after the desired contact period (18 h).

2.3.4. Antiallergic studies

The antiallergic tests were performed according to the 'appraisal of the safety of chemicals in foods, drugs and cosmetic' by the staff of the division of pharmacology (FDA) and the OECD-Guidelines (No. 4.4 on May 1992). Interpretation was made according to the code of Draize1959, ISO 10993-10:1995 and Get Stoff V 2000 (Regulations for German dangerous classification).

The primary skin irritation has been measured with the patch test method on the shaved skin of albino rabbits. In this method, 500 mg of textile sample with dimensions of 45 × 45 mm was applied. The samples have been wetted by spraying 2 ml water and covered with 12 × 12 cm gauze squares on the skin. The gauze patches were fixed with adhesive plaster. The experimental rabbits were kept in a small cage for an hour

and then returned to their homes. After 4 h, the punch was removed. After 1 h of punch removing, the local reactions were evaluated according to Draize procedure. The evaluations were repeated every 24 h after punch removal and mean value were scored. The final control of localizations has been carried out after 7 days. For the reproduction of the values, all experiments were repeated again.

2.3.5. Mechanical and optical tests

Investigation of the mechanical properties (tensile strength and elongation) was carried out according to DIN 53857 using a Zwick 1445 and ISIRI 1667, 2794 using the Lenzing testing System.

The UV barrier of the modified textile products was measured according to AS/NZS 4399:1996 – sun protective clothing – evaluation and classification using a Cary 50 solar screen transmission spectrophotometer and expressed in term of the ultraviolet protection factor (UPF), that was evaluated on the basis of UV transmittance measurements by the spectrophotometric method using a double beam V-550 type Jasco UV-Vis spectrophotometer equipped with an integrating head. The UPF value was calculated automatically according to (Equation (2)) and classified according to Table 1.

$$UPF = \frac{\int_{\lambda=290}^{\lambda=400} E(\lambda)\epsilon(\lambda)d(\lambda)}{\int_{\lambda=290}^{\lambda=400} E(\lambda)\epsilon(\lambda)T(\lambda)d(\lambda)} \quad (2)$$

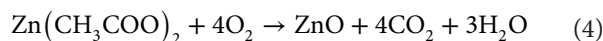
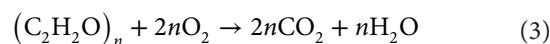
where, $E(\lambda)$, $\epsilon(\lambda)$, $\Delta \lambda$ and $T(\lambda)$ are the solar irradiance, the erythema action spectrum (Ciritia of the harmfulness of UV radiation for human skin, connected with the wavelength, the erythema effectiveness is highest at about 290–310 nm), the wavelength interval of the measurements and, the spectral transmittance at wavelength λ , respectively.

3. Results and discussion

3.1. Synthesis of ZnO nanorods

Synthesis of ZnO nanorods was done using PVA-based sol-gel Pyrolysis method which was introduced in our previous work (Karami, Aminifar, Tavallali, & Namdar, 2010). In the PVA-based gel combustion method, the gel network rigidity controls the morphology and particle size of the synthesized zinc oxide nanostructures. To make uniform ZnO Nanorods, zinc acetate molecules were homogeneously dispersed among a polymeric network of the gel. Because of gel network rigidity, the dispersed molecules in the gel network cannot alter their positions. Therefore, during the Pyrolysis of the gel, outer layers of gel were burnt, and zinc acetate contents of the burnt layer calcinated into ZnO. The following reaction was performed

during Pyrolysis (Equation (3)) and calcination (Equation (4)) processes:



In this method, the amount of Zn $(CH_3CCO)_2$, PVA, the type and composition of solvent, pH, temperature and time of paralysis, and the type and amount of additives are the effective parameters of the synthesis which, their amounts are changed and optimized by the 'one at a time' method. Several synthesizers were performed to obtain optimum conditions for synthesizing uniform zinc oxide Nanorods. The obtained experimental results showed that the optimum conditions to synthesize uniform ZnO Nanorods are including 6 %wt zinc acetate, 3 %wt PVA, mixed solvent ethanol-water (70:30), 400 °C pyrolysis temperature, 5 %wt PVP as additive, 4 h pyrolysis time, and pH solution of 8. Figure 1 shows the SEM and TEM images of the ZnO sample synthesized at the optimized conditions. As it can be seen in Figure 1, the sample contains uniform nanorods with 60 nm average diameter and 1000 nm average lengths.

3.2. Characterization of ZnO/PP nanocomposite fibers

The synthesized ZnO nanorods were mixed with the melted PP and the ZnO/PP nanofibers were produced in pilot scale according to Section 2.3.2. The produced sample was initially analyzed by X-ray diffraction spectroscopy (XRD). The XRD patterns confirmed the presence of ZnO in the PP matrix. Finally, the sample was studied by SEM and TEM images. Figure 2 shows the SEM and TEM images of the preparation ZnO/PP nanocomposite fibers. SEM image (Figure 2(A)) shows that the surface of the prepared nanocomposite fiber is uniform. TEM image (Figure 2(B)) obviously shows the presence of ZnO nanorods in the PP matrix.

3.3. Antibacterial activity

The results of the qualitative antibacterial assessment by the agar diffusion show that the fabric treated with ZnO Nanorods has a maximum inhibitory effect against *E. coli* and *S. aureus*. Figure 3 shows the photographs of *E. coli* cultivation media for virgin fibers, fibers treated with micrometer sized ZnO, and the fiber treated with ZnO nanorods. As it can be seen in Figure 3, The presence of zinc oxide reduces the growth of bacteria colonies. Antibacterial activity of ZnO nanorods is much better than micrometer-sized Zn. The summary of the obtained results were shown in Table 2. The presented data show that the fibers treated with ZnO nanorods has more antibacterial activities against all types of bacteria than both virgin fabric and fabric treated with micrometer-sized ZnO.

The antibacterial activity of the ZnO particles was studied. It seems that active oxygen species generated by ZnO particles

Table 1. UPF classification system according to AS/NZS 4399 (1996) standard [18].

UPF range	UVR protection category	Effective UVR transmission%	UPF rating
15–24	Good protection	6.7–4.2	15, 20
25–39	Very good protection	4.1–2.6	25, 30, 35
>40	Excellent protection	≤2.5	40, 45, 50, 50+

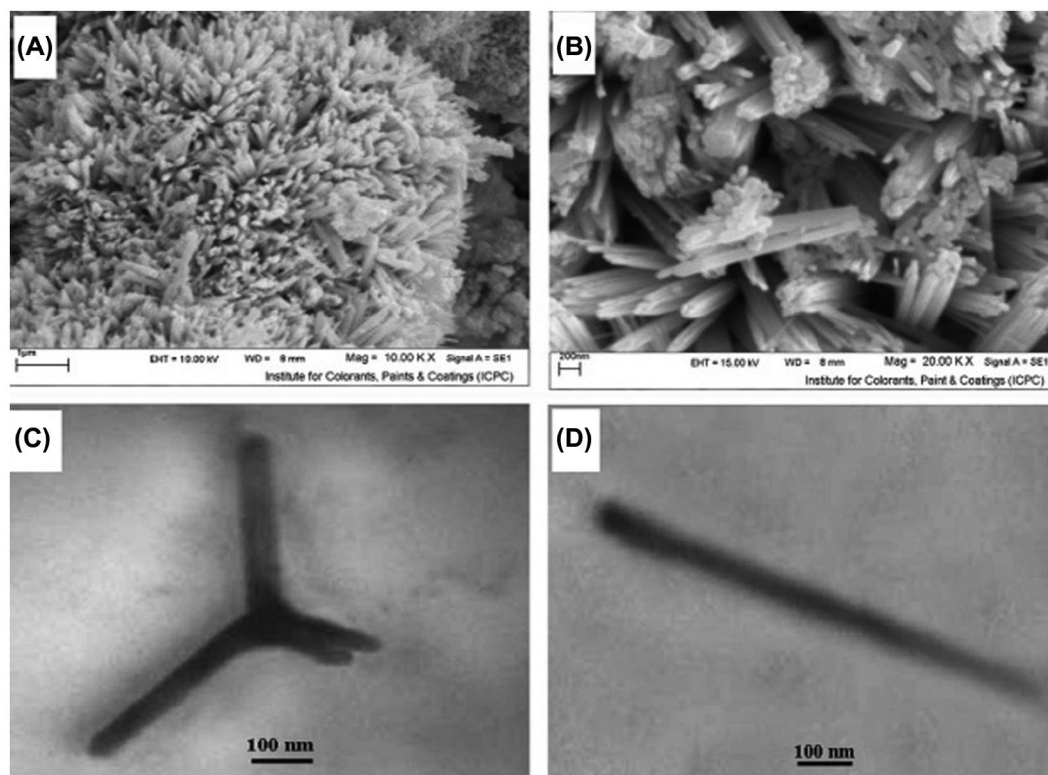


Figure 1. SEM (A and B) and TEM (C and D) images of the synthesized ZnO nanorods.

could be a main factor of the antibacterial mechanism, although there is no direct evidence for this conclusion.

The presence of active oxygen species has been detected. (Yamamoto, Sawai, & Sasamoto, 2000). It has already been proved that both nanometer-sized and micrometer-sized ZnO suspensions are active in inhibiting the bacterial growth. The previous reports show that ZnO nanoparticles have a much higher activity than the micrometer-sized ZnO (Zhang et al., 2010). The mentioned previous results are confirmed by the results of the current study.

The quantitative bacterial reduction was studied by the percentage reduction test and the results were shown in the Table 3. The results of this percentage reduction test correspond with that of the agar diffusion and parallel streak method. The ZnO Nanorods-treated fibers showed a maximum percentage reduction of 95% for *E. coli* followed by 86.5% for *S. aureus*. The fiber treated with micrometer-sized ZnO expressed comparatively a lower percentage of reduction. The fiber without any treatment (control sample) has negative values for the percentage reduction test because the final number of cells is higher than the initial number of cells as it has no antibacterial activity.

3.4. Antiallergic effects

The results of antiallergic tests of ZnO/PP fibers in rabbit skin, according to the OECD-guidelines No. 4.4. Based on the obtained results, there is not any skin exema or sensible allergic effects on the skin of examining rabbits during 7 days. The animals gained weight in normal range and no other signs in their behaviors. Therefore, there is not any objection for the contact of normal human skin with ZnO-PP nanocomposite fiber and non-woven fabrics.

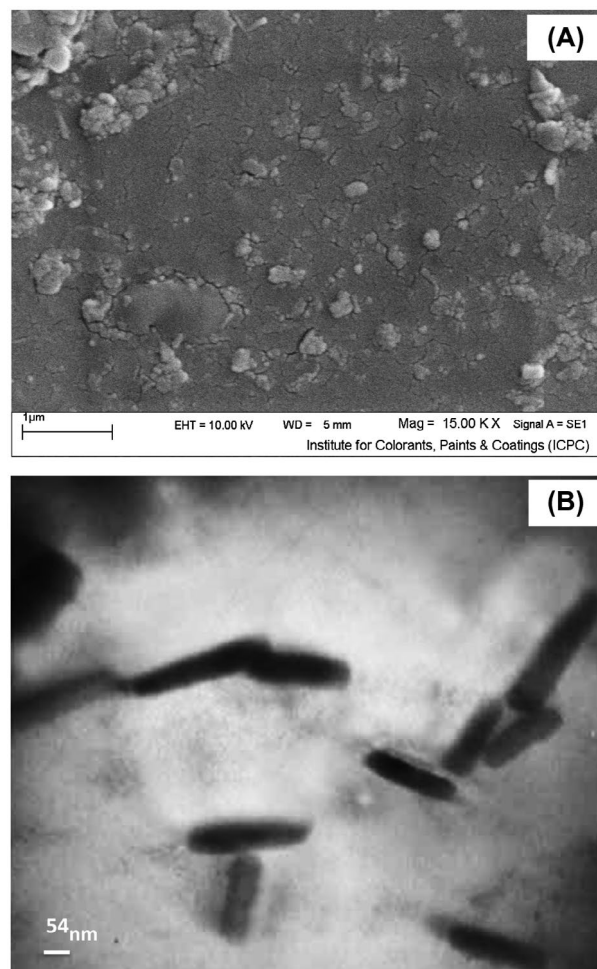


Figure 2. SEM (A) and TEM (B) images of the prepared ZnO/PP nanocomposite.

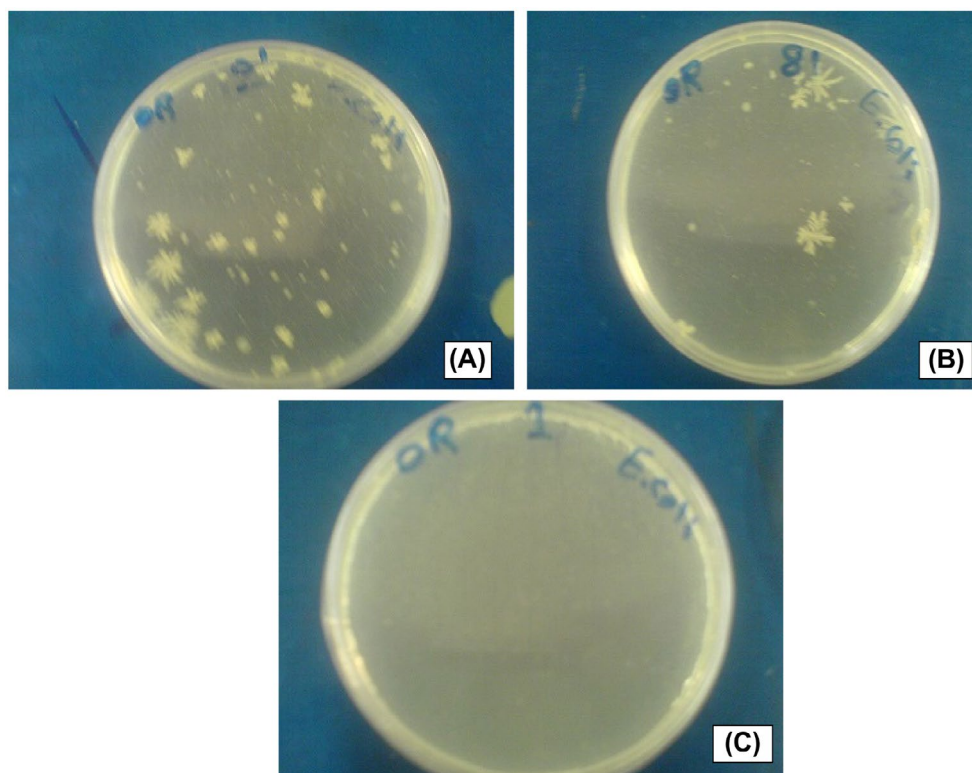


Figure 3. Antibacterial activities of untreated fibers (A), treated fiber with micrometer-sized ZnO (B) and treated fibers with ZnO nanorods (C).

Table 2. Antibacterial assessment By agar diffusion method.

Fabre treated	Organism	Zone of inhibition (in cm)				
		Trail 1	Trail 2	Trail 3	Mean	Range
Fibers treated with ZnO nanorods	<i>E. coli</i>	4.8	5.4	5.4	5.2	0.5
	<i>S. aureus</i>	3.4	3.4	3.1	3.3	0.20
Fibers treated with micrometer sized ZnO	<i>E. coli</i>	3.1	2.8	2	2.6	0.50
	<i>S. aureus</i>	1.8	1.9	1.9	1.8	0.3
Virgin fibers	<i>E. coli</i>	0	0	0	0	0
	<i>S. aureus</i>	0	0	0	0	0

Table 3. Results of the reduction tests (number of colonies) for three bacteria in contact with three types of fibers.

Fabric tested	Organism	Initial number of bacterial cells/ml	Final number of bacterial cells/ml	% reduction
Fibers treated with ZnO nanorods	<i>E. coli</i>	6×10^6	3×10^5	95
	<i>S. aureus</i>	6×10^6	8.4×10^5	86
Fibers treated with micrometer-sized ZnO	<i>E. coli</i>	6×10^6	3×10^6	50
	<i>S. aureus</i>	6×10^6	3.78×10^6	37
Untreated fibers	<i>E. coli</i>	6×10^6	$>6 \times 10^6$	0
	<i>S. aureus</i>	6×10^6	$>6 \times 10^6$	0

3.5. Physical, mechanical, and optical properties

First, the fiber samples were tested in electromagnetic field for the determination of titer_(detox). The other properties such as tenacity and elongation were tested by a fixed force tearing and the results were summarized in Table 4. The results of tenacity show that the nanocomposite fibers have higher tenacity than two other fibers.

One of the most destructive processes in PP polymer is photodegradation. In this study, the fiber resistance against UV light were investigated for three samples of virgin fibers, ZnO/PP nanocomposite fiber and micrometer-sized ZnO/PP composite fiber. The tenacity and elongation tests also were done after UV

irradiation (Table 4). The effects of UV irradiation were investigated by stress-strain curves (Figure 4). As it can be seen in Figure 4, sample A (nanocomposite fiber) have higher tenacity in comparison with other fibers after UV light irradiation.

3.6. The UV blocking test

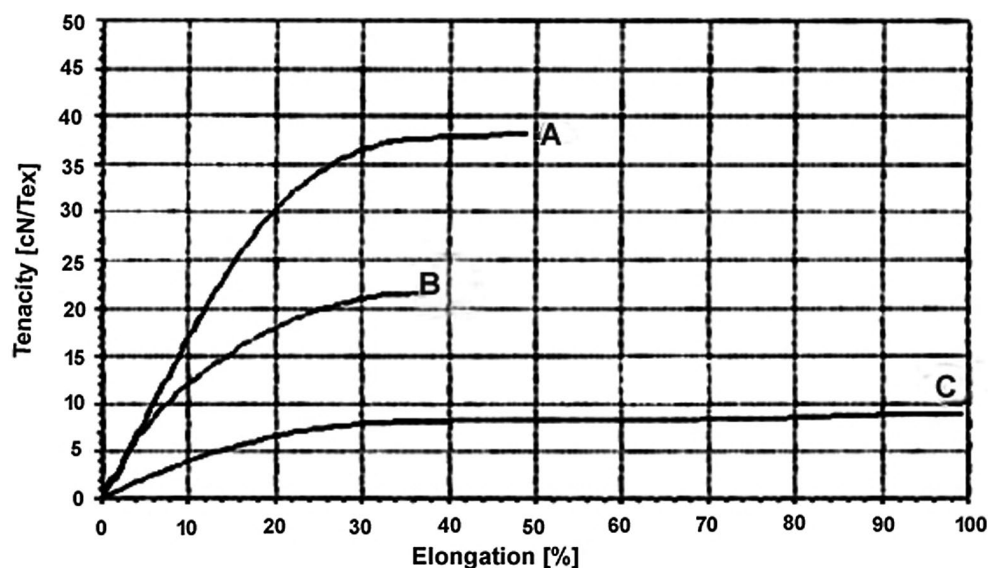
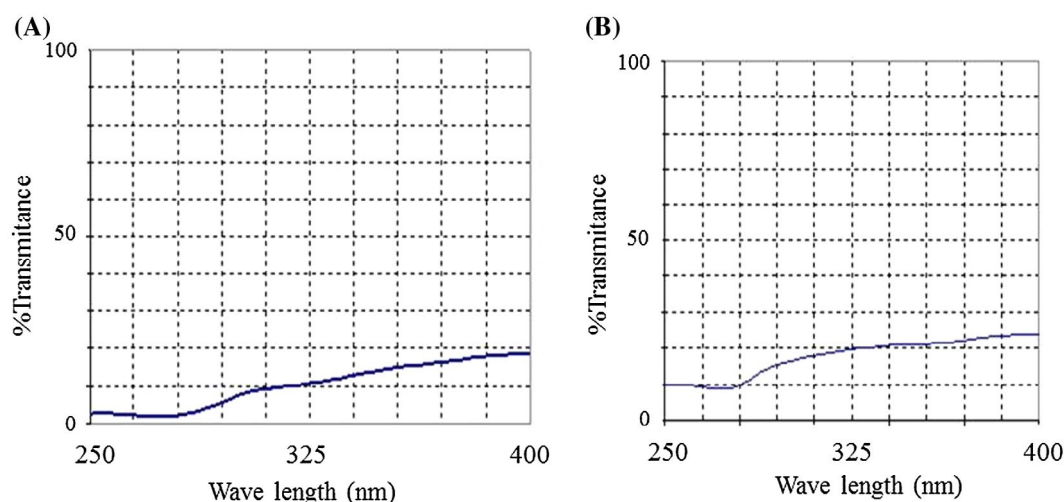
The fibers and non-woven fabrics can be used for the production of clothing. Therefore, the UV light blocking property is a main factor in these fibers. In this method, the amount of UV light transmission through the textile is determined by the

Table 4. The special physical properties of the three fiber samples before and after UV irradiation.

Sample fiber	Titer (dtex) CV%	Tenacity (CN/TEX) CV%	Elonga- tion (%) CV%	Force (CN) CV%
ZnO nanorods/PP nanocomposite fiber before lightening	11.65 14.4	39.3 9.9	76.5 32.2	45.42 11.4
ZnO nanorods/PP nanocomposite fiber after lightening	10.65 14.4	38.5 10	49.3 30	44.1 12
Micrometer-sized ZnO/ PP composite fiber before lightening	10.67 13.1	28.2 14.6	57.8 26.3	40.42 21.6
Micrometer-sized ZnO/ PP composite fiber after lightening	10.67 13.1	22.3 14	33.5 27	12 20
Virgin fiber before lightening	10.11 5.6	31.2 10.3	63.3 26.8	31.43 8.7
Virgin fiber after lightening	10.11 5.6	9.8 12	100 25	32 10

UV-Vis spectroscopy. In practice, a thin woven sample is inserted between two wood washers and, the source light was passed from the center of the sample. The comparison of UV-spectra of fabrics treated with ZnO nanorods and untreated fabrics were shown in Figure 5. The results shows decrease in the transmission of UV radiation in curve A for ZnO-PP Nanocomposite in limit wave length 250–359.

The prepared ZnO- pp nanocomposite was analyzed by powder XRD technique (Figure 6). The presented XRD patterns shows the presence of ZnO in polymeric matrix. In the standard of AS/NZS 4399:196, with respect to the light protection efficiency, the fibers are classified. In this standard, the fibers are divided into three classes including good protection, very good protection and excellent protection based on UPF. Based on the obtained results, nanocomposite fiber (fibers treated with ZnO nanorods) and nanocomposite nonwoven fabrics have excellent protection in the wavelength of 250–300 nm (Figure 7).

**Figure 4.** Stress-strain curve after UV light irradiation: (A) ZnO nanorods/PP Nanocomposite fiber, (B) Micrometer-sized ZnO/PP composite fiber and (C) Virgin fiber.**Figure 5.** UV spectra for the ZnO nanorods/PP nanocomposite fiber (A) and untreated PP fiber (B).

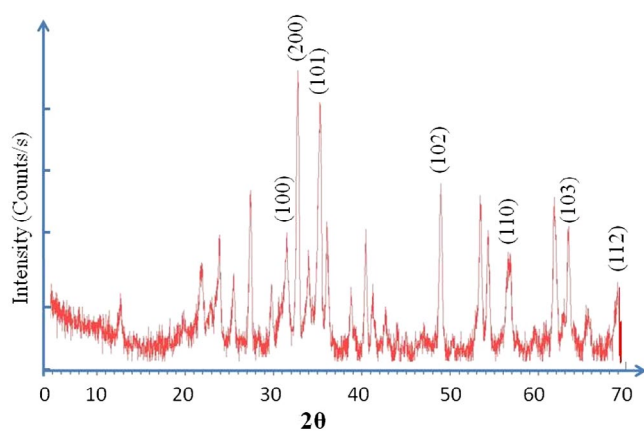


Figure 6. XRD patterns of the prepared ZnO-PP nanocomposite.

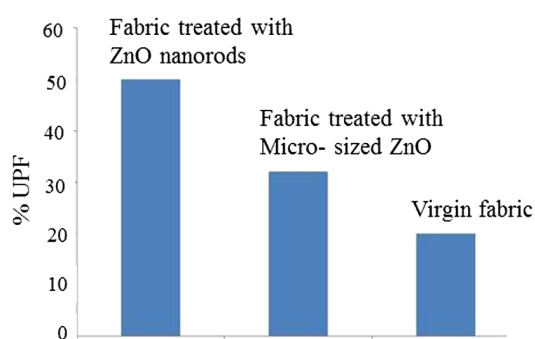


Figure 7. The light protection efficiencies of virgin fabric, fabric treated with ZnO nanoparticles and fabric treated with micro-sized ZnO.

4. Conclusions

It is concluded that the presented method is a new, selective, and template-free method for the synthesis of ZnO nanorods based on the PVA gel pyrolysis method. The ZnO/PP nanocomposite were prepared by extrusion method. The prepared nanofibers and nonwoven fabrics shows excellent antibacterial activity against *E. coli* and *S. aureus*, high UV blocking performance, high mechanical strength and no allergic effects. Based on the presented results, ZnO nanorods can be used in production of polymeric fibers, nonwoven fabrics and any other textiles. These fibers and fabrics are most suitable for defense clothing like gloves and T-shirts to avoid cross infections. It was found that the amount of ultraviolet radiation absorbed by these particles (nanorods) was much greater than the other conventional materials used, they can be therefore, a superior alternative. This technology can be further extended to polyester and other fibers.

Disclosure statement

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